

# Entity Relationship Model (ER Model)

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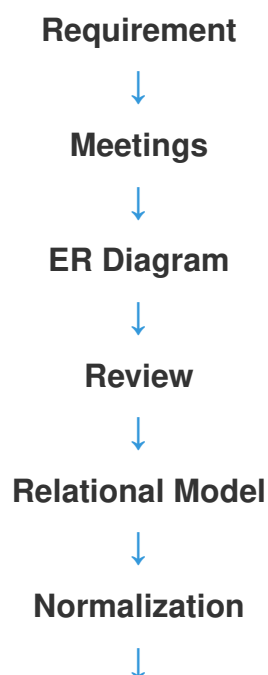
An Entity Relationship Model (ER Model) is a high-level conceptual representation of the data, entities, attributes, and relationships in a database. It describes:

- what data exists
- how data is connected
- what rules govern the data

## **ER Modeling helps:**

- reduce redundancy
- avoid anomalies
- identify relationships
- improve normalization
- simplify implementation

## **Typical Workflow**



## SQL Tables



Application (testing, deployment, maintenance)

## Core Components

ER model is all about entity, relationship, attributes, constraints.

## Symbols Used in ER Diagrams

The classic Chen notation uses these symbols:

| Symbol               | Meaning                  |
|----------------------|--------------------------|
| Rectangle            | Entity                   |
| Double Rectangle     | Weak Entity              |
| Diamond              | Relationship             |
| Double Diamond       | Identifying Relationship |
| Oval                 | Attribute                |
| Double Oval          | Multivalued Attribute    |
| Dashed Oval          | Derived Attribute        |
| Underlined Attribute | Key Attribute            |

## Design Phases

### 1. Conceptual Design

### 2. Logical Design

Now convert the conceptual model into relational structures.

Example:

- **Student:** StudentID, Name
- **Course:** CourseID, Title
- **Enrollment:** StudentID, CourseID

Primary keys and foreign keys are decided here.

### 3. Physical Design

Finally, decide how the database will be implemented. Examples:

- data types
- indexes
- partitions
- storage engine

SQL statements are written in this phase.